

How In-Context Learning Reorganizes Representations: the Role of Domain, Ordering and Graph Structure

Giacomo Amerio, Valeria De Stasio, Giovanni Lucarelli

github.com/Project-Hindenburg/hico

1 Introduction

One of the most striking capabilities of Large Language Models (LLMs) is their ability to adapt to novel tasks without any parameter update, relying solely on information provided in the context window. This phenomenon, termed In-Context Learning (ICL), has given rise to much interest in the ML community. Yet despite considerable progress, a fundamental question remains under explored: to what extent does the *structure* of the context, both the topology of the relationships being communicated and the semantic nature of the tokens present in the relationships, determine how a model internally reorganizes its representations in response?

A recent perspective on how context reshapes representations comes from [3]. In their graph tracing paradigm, a model is shown tokens from a random walk over a graph whose nodes are labelled with semantically uncorrelated concepts (*e.g.*, apple, bird and code) arranged in an arbitrary geometric structure such as a square grid, ring, or hexagonal lattice. As context accumulates, the model’s internal representations undergo a phase-transition like reorganization: their geometry in activation space comes to mirror the graph structure underlying the data generating process. The visualization of the first two principal components of the mean token representations clearly shows the latent graph structure, and this reorganization is accompanied by a corresponding improvement in the model’s ability to predict valid next nodes. They quantitatively characterized this transition by observing a drop in the Dirichlet energy and a coherent increase in the next token prediction accuracy. The only experiment involving semantically structured tokens considers the days of the week arranged on a circular graph (tokens whose embeddings already exhibit a circular geometry in pretrained models [1]). In this case, the model does not recover the imposed graph structure in the leading principal components, and the authors interpret this result as evidence that strong pretrained priors can inhibit contextual geometric reorganization.

This interpretation leaves open two questions that we believe are fundamental. First, is the resistance observed with days of the week due to the *strength* of the pretrained prior, or is it due to the specific *geometric relationship* between the pretrained embedding structure and the target graph topology? Second, are next-token prediction accuracy and geometric reorganization necessarily coupled, or can a model learn to solve the graph-tracing task without constructing a geometric representation of the graph?

This work addresses both questions through a series of experiments across combinations of different graph topologies (tree, circle, grid) and different semantic domains (uncorrelated words, days of the week, natural numbers). We further distinguish between ordered and unordered assignments of words to graph nodes whenever the semantic domain admits a notion of order.

Our results show that both graph topology and semantic domain play an important role, and, crucially, that their interaction reveals a richer picture of in-context representational reorganization than previous work has suggested. Regarding the first question, we find evidence of a positive interference effect when the target graph is *aligned* with pretrained semantic. Regarding the second question, our experiments show that geometric reorganization is not always tightly coupled with predictive performance: as the context window grows, models can improve at the graph-tracing task even when the expected geometric restructuring is weak or absent.

2 Methods

Our experiments were conducted using the base version of Llama-3.1-8B and proceeded as follows for each domain:

1. Verify that the LLM tokenizer maps the selected items to single tokens.
2. Generate prompt files as random walks over the chosen graph structures.
3. Feed each prompt to the LLM and save both the embeddings from specific layers and the next-token prediction probabilities at various positions in the prompt.
4. Analyze the extracted representations.

2.1 Dataset generation

For each combination of graph structure (*i.e.*, tree, grid, and circle) and token domain (uncorrelated words, days, and numbers), we constructed prompts as single-line strings generated by a random walk of length $l = 2000$ on the corresponding graph. For the domains of days and numbers on circle and grid structures, we additionally considered two possible arrangements: an increasing ordering, which we denote as *semantic order* (clockwise for the circle and row-wise for the grid), and a random arrangement over the same structure, which we denote as *shuffled order*. Samples were generated by first selecting a random starting node and then traversing the graph through a random walk. At each step, the transition to a neighboring node was sampled uniformly from the set of adjacent nodes.

Each generated prompt was fed to the LLM, and token embeddings were extracted from the residual stream ($d = 4096$) of the final transformer layer, namely layer 32.

2.2 Tokens Selection

We chose three different domains of words: 16 semantically uncorrelated English words, the days of the week, and integer numbers in the range 1-16. Note that in some cases we had to reduce the domain to better fit some structures (*e.g.* using only 15 numbers to fit a full binary tree of height three).

To select uncorrelated words, we began with a large list of English words (about 350K), keeping only lowercase words made entirely of alphabetic characters. We then prefixed each word with a space and passed it through the tokenizer, retaining only those that corresponded to a single token. Next, we extracted the embedding of each word from layer 0. Starting from one randomly chosen word, we iteratively added new words by selecting those with the lowest cosine similarity to the words already chosen, continuing until we obtained the desired number of words.

2.3 Assessment Metrics

Dirichlet Energy. Following [3], in order to quantify the smoothness of the learned representations over the graph, we compute the Dirichlet energy of the node embeddings. For a graph $G = (V, E)$ with Laplacian $L = D - A$ and node embedding matrix $X \in \mathbb{R}^{|V| \times d}$, where x_i denotes the embedding of node i , the Dirichlet energy can be written as

$$E(X, L) = \text{Tr}(X^\top LX) = \frac{1}{2} \sum_{i,j} A_{ij} \|x_i - x_j\|_2^2$$

and its normalized version, removing the dependence on the global scale of the embeddings, can be written as:

$$E_{\text{norm}}(X, L) = \frac{E(X, L)}{\|X\|_F^2}$$

This quantity is small when neighbouring nodes have similar representations, and large when embeddings vary sharply across adjacent nodes. In our procedure, for each context window, we aggregate the last $k = 5$ token occurrences into centroid embeddings, center their representations, and project them onto a two-dimensional subspace obtained by selecting the first two principal components of the 4096-dimensional embedding space.

Graph Tracing Loss To quantify the ability of the LLM in the Next Token Prediction (NTP) task as a function of the context length, following [3], we define the *graph tracing loss* as

$$\mathcal{J}_t = -\log \sum_{v \in \mathcal{N}(x_t)} p_\theta(x_{t+1} = v \mid x_{\leq t}),$$

where $p_\theta(\cdot \mid x_{\leq t})$ denotes the model next-token distribution after the prefix $x_1, \dots, x_t$, and $\mathcal{N}(x_t)$ is the set of vocabulary tokens corresponding to the neighbours of x_t in the chosen graph (after token normalization/decoding). In practice, we approximate the sum by restricting it to the model top- k predictions at time t .

Note that this metric does not quantify how well the graph structures are reconstructed in the embedding space but only the ability to improve in the next token prediction task as the context length increases. Together the two metrics reveals two different phenomena: the geometrical rearranging in matching the underlying graph structure and the NTP ability of the model.

Agglomerative Hierarchical Clustering. For prompts coming from tree structured graphs, we also performed agglomerative hierarchical clustering (using Ward’s distance) on the centroids of the representations in order to assess whether and at which degree the contextualized embeddings encoded the hierarchies of the tree.

3 Results

3.1 Uncorrelated Words

Using uncorrelated words, we find across all graph structures, results that are consistent with [3] (Figure 1). Specifically, in the absence of *interference* between the semanticity of the tokens (being the words uncorrelated) and the underlying graph structure, the representations can be reorganized in terms of the first two principal components.

The tree dendrogram (Figure 2) shows that the embeddings’ rearrangement enables the agglomerative clustering procedure to correctly group tokens belonging to the same tree depth. However, it fails to recover the full global hierarchy, *i.e.*, it does not merge tokens across increasing distances in a way that reflects decreasing depth in the tree.

In all these cases, we observe that both the loss and the Dirichlet energy decrease as the context length grows (Figure 3), indicating that the model is able to learn the new task in context as well as correctly reorganizing its internal representations according to the latent graph structure. In these settings, our experiments point towards the same direction claimed by [3], this geometric rearrangement may be the crucial factor that allows the model to solve the NTP task.

3.2 Days

Circle. On the circle structure, we recover the results of [3] (Figure 4). For ordered days, the circular structure is recovered in the first two principal components. For shuffled days, the semantic order emerges in the first two principal components, while the structural order appears in the second and third components. This may happen because the direction of largest variance is dominated by the semantic prior [3].

Grid. On the grid structure, the interference between the tokens domain and the graph structure is larger because of the vertical connections (e.g., Monday–Thursday), and it increases further when the semantic order is disrupted by shuffling. In both cases, geometric reorganization is never fully achieved, although the model is still able to solve the NTP task effectively (Figure 5). This suggests that the model is performing in-context learning not through the reorganization of its embeddings, but rather by relying on pattern-matching abilities [2], without substantially reshaping the representation space.

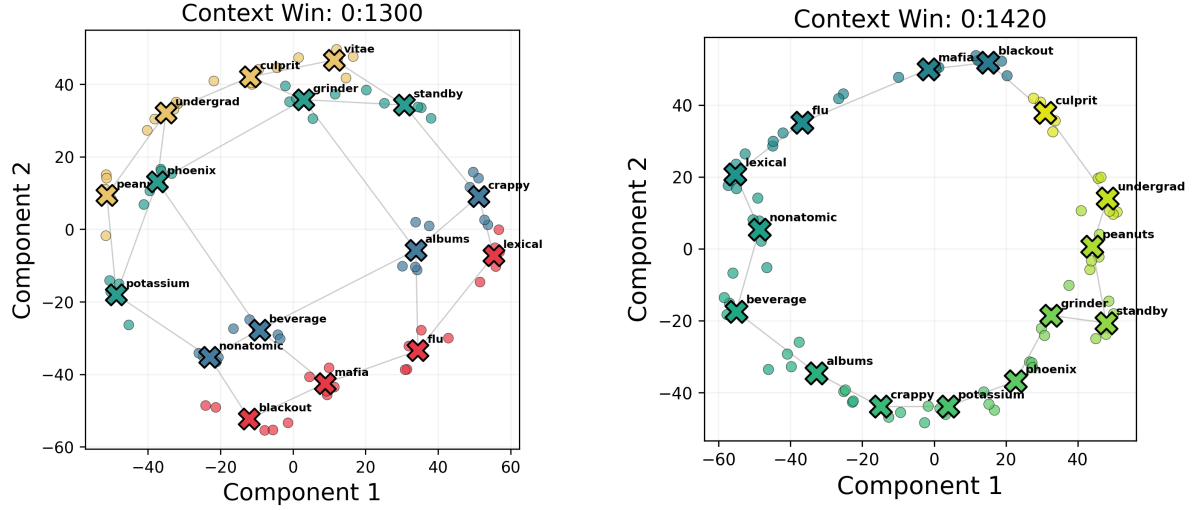


Figure 1: Uncorrelated words. First two principal components of the embeddings corresponding to the last five occurrences of each word in the optimal context length; the $\times$ marker denotes the centroid of each word. Left: grid structure. Right: Circle structure.

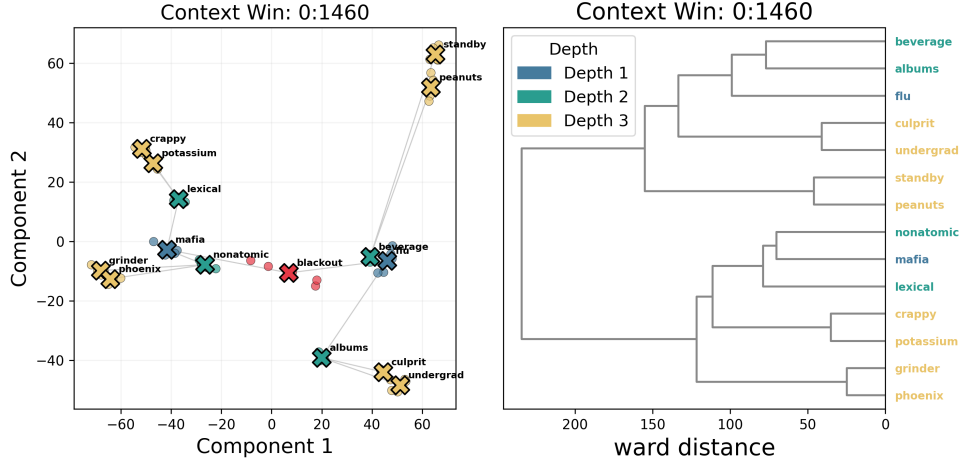


Figure 2: Uncorrelated words on a binary tree structure. Left: first two principal components of the embeddings. Right: dendrogram obtained through hierarchical agglomerative clustering.

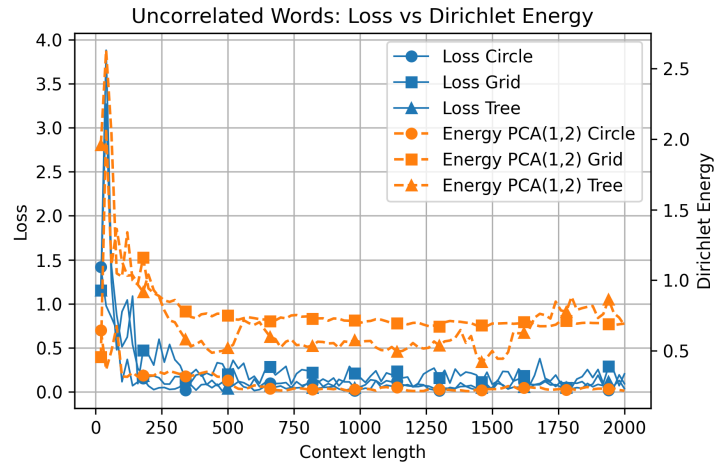


Figure 3: Uncorrelated words losses (in blue) and energies (orange) for the three graph structures.

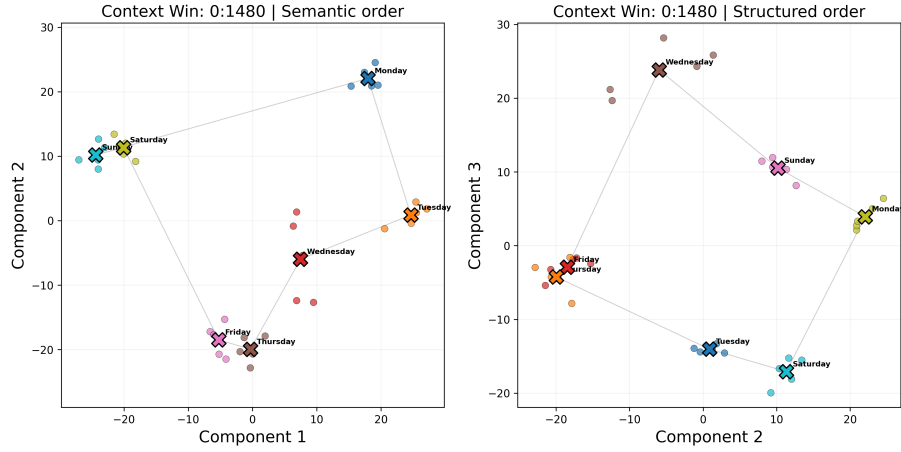


Figure 4: Shuffled days on circle. Left: first two principal components encode the semantically ordered circle. Right: second and third principal components encode the shuffled ordered circle.

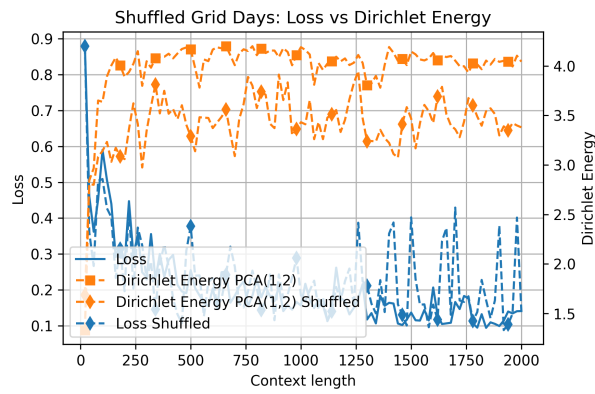


Figure 5: Days on grid (2×3), ordered and shuffled. Loss goes to zero for ordered and shuffled whereas the Dirichlet energies reach a high plateau, meaning that the embeddings do not match the respective graphs.

Tree. For days, there is no natural notion of order on a binary tree structure, so we refer to this setting as an unordered tree. The tree structure is correctly recovered and is already identifiable in the first two principal components (PCs)¹. The Dirichlet energies reach a plateau at the same point as the loss, suggesting that, in this case, a structural reorganization is indeed happening and it may be helpful for the NTP task (Figure 6, 7).

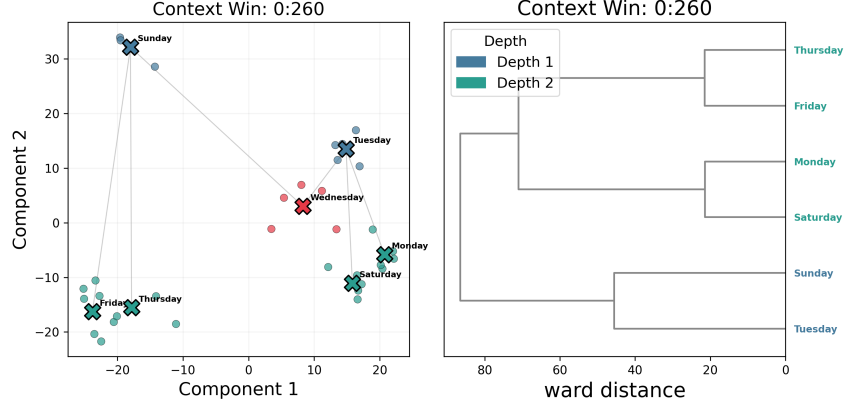


Figure 6: Days on binary tree. Top: first and second components for the optimal energy value. Bottom: second and third components for the optimal energy value.

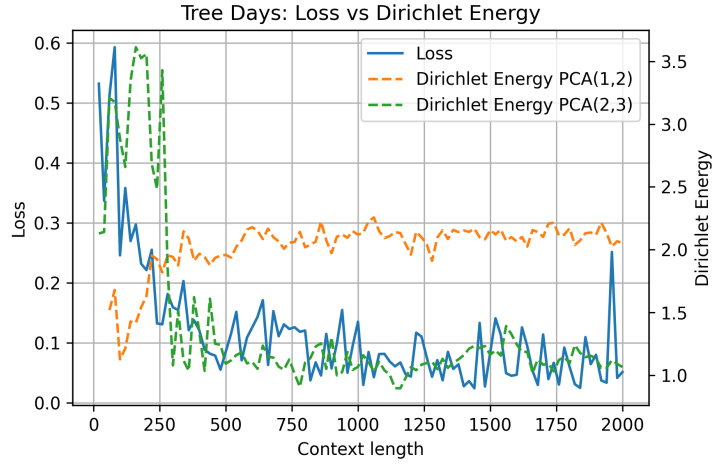


Figure 7: Days on binary tree. Loss (blue) and energy relative to first and second components (orange), and second and third components (green).

3.3 Numbers

Tree. As in the case of days, for the tree graph structure we are able to identify the tree geometry in the first two principal components, as also confirmed by the dendrograms (and, again, this is more clearly visible in the second and third components). As before, there is no natural notion of order on a binary tree. Nevertheless, the fact that both the Dirichlet energy and the loss reach a plateau suggests that the model solves the NTP task while simultaneously rearranging the representations to reflect the graph structure.

Semantic Order. For circle and grid structures, where a natural ordering can be defined, we observe that the geometry is recovered in the first two principal components. As in the experiments with days, the circle exhibits a lower Dirichlet energy than the grid (Figure 8), suggesting a better alignment between the graph structure and the embedding geometry. We interpret this difference as a consequence of the

¹Although the lowest Dirichlet energy is obtained by the second and third principal components, we interpret this as evidence that a tree can be arranged in multiple equivalent ways in two-dimensional space, whereas circles and grids are subject to stronger structural constraints.

slightly stronger constraints imposed by the grid, which may make it less compatible with the embeddings of ordered numbers.

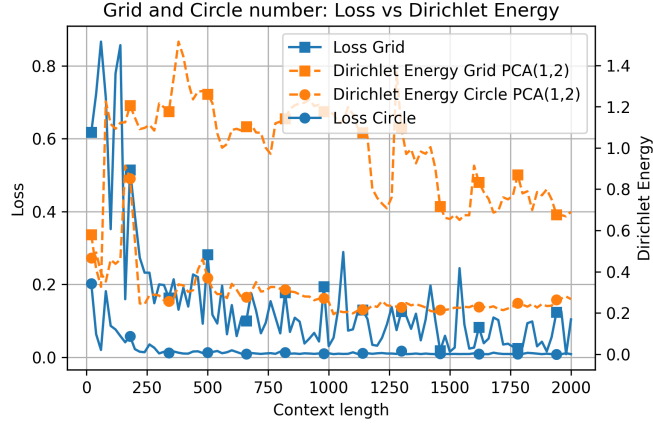


Figure 8: Ordered numbers on circle and grid. Losses (blue) and energies (orange). We observe rearrangement in both structures, *i.e.*, low energy.

Shuffled Order. Shuffling the numbers increases the level of interference and, for both the grid and the circle, leads to a decoupling between the Dirichlet energy and the loss behaviors. In the grid case, we do not observe any geometric arrangement consistent with the underlying graph, analogously to what we found for shuffled days on the grid (Figure 9). Despite this, the model remains able to solve the NTP task effectively, suggesting that in-context learning can still occur without a corresponding geometric reorganization.

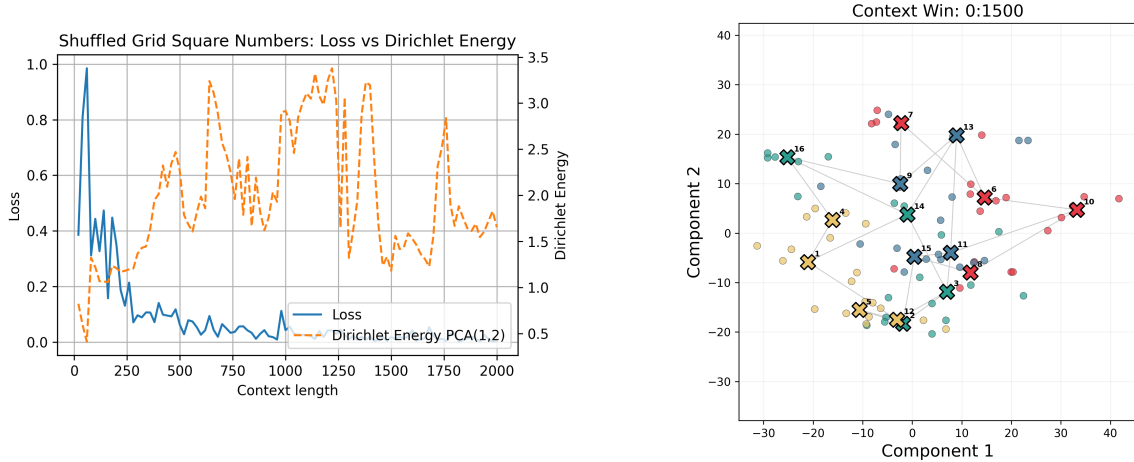


Figure 9: Shuffled numbers on grid. Left: loss (blue) and energy (orange). Right: first and second principal components of the embeddings do not encode the grid structure.

The shuffled circle case is richer than the shuffled grid one. We are able to identify the correct geometric structure only within a specific context length, namely $l = 1480$ (Figure 10). Before this point, the system appears to go through a transient phase; around this context length, it reaches an equilibrium corresponding to the minimum Dirichlet energy. For larger context, however, the Dirichlet energy increases again, indicating a further rearrangement of the embeddings even as the loss continues to decrease (Figure 11). Interestingly, we observe that information about the order is consistently encoded in the first principal component, while it alternates between the second and third components as the context length increases. This suggests that the amount and organization of the variance needed to represent the new order change across context lengths, with information effectively being redistributed between the second and third principal components.

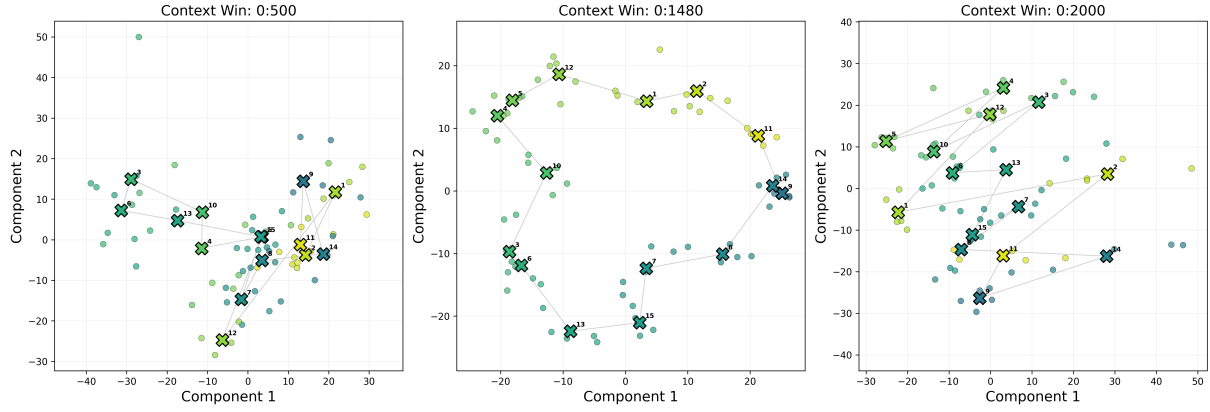


Figure 10: Shuffled numbers on circle. First two principal components for increasing context lengths from left to right.

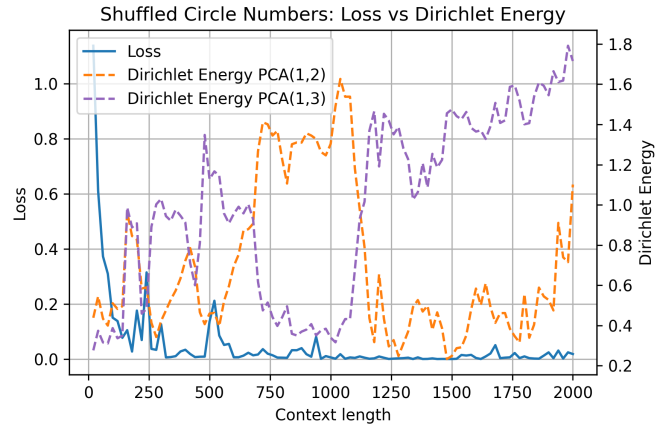


Figure 11: Shuffled numbers on circle. Loss (blue) and energy relative to first and second components (orange), and first and third components (violet).

4 Discussion

Our experiments systematically vary topology, token domain and arrangement order, revealing a picture of in-context representational reorganization that is more structured (and surprising!) than prior work has suggested. Two findings stand at the center of this picture.

The interference principle. The variable that best predicts whether geometric reorganization succeeds is *not* the strength of the pretrained prior as hinted by [3], but its geometric relationship to the target graph topology. The days of the week provide the cleanest proof of this: the same tokens, carrying the same pretrained circular geometry [1], are placed on three different structures. On a tree, reorganization succeeds in PC1-PC2, so the pretrained circle is overwritten. On a shuffled circle, the in-context structure is displaced to PC2-PC3, reproducing the finding of [3]. On a grid, geometric reorganization fails entirely.

It is thus easy to see that, if prior strength was the dominant variable, all three conditions would behave similarly, because the prior is equal in each. Instead, the outcomes highlight geometric compatibility as the best explanation for this behaviour. A tree has multiple possible linear representations in two dimensions (it is thus more *flexible* than the other two topologies), its representational requirements are possibly orthogonal to those of the pretrained circle, so no collision occurs. A shuffled circle needs the same planar dimensions that the pretrained circle occupies, but with an incompatible node ordering, hence the two structures compete for the same subspace. Finally, in both the ordered and the shuffled days grids the model is not able to reorganizes its internal structure to mimic the graph due to the strong incompatibility between the natural topology and the target graph one.

The numbers experiments remark the role of geometric interference in IC representation reorganization. Ordered numbers succeed on both circles and grids because the pretrained number line is compatible with both layouts (though the circle achieves lower Dirichlet energy than the grid, reflecting the greater affinity between a line and a circle than between a line and a two dimensional lattice). Shuffled numbers on a grid never reorganize, and shuffled numbers on a circle produce only a transient circular structure that appears at intermediate context lengths before dissolving. Hence, shuffling destroys the geometric compatibilities.

Trees are immune to interference. They are recovered in every domain and every ordering condition, we attribute this to the flexible nature of tree topology. As confirmed by the dendrograms, the reorganization is such that preserves the local neighbourhood, *i.e.*, it correctly reorganizes nodes of the same depth as neighbors, although it does not capture parent-child relationships.

Two separable modes of in-context learning. The interference principle tells us when geometric reorganization fails, but in every such condition (*i.e.*, shuffled numbers on a grid, days on a grid, shuffled days on a circle) the graph tracing loss *still drops to zero*. The model solves the task without building a geometric representation of the latent structure.

This dissociation establishes that ICL may operate through at least two modes. **Pattern matching** learns local transition statistics [2]: given that the current token is x_t , which tokens have previously appeared adjacent to x_t in the context? This is sufficient for task accuracy and is robust across all conditions. We call the second mode **Geometric matching**. In this scenario, the model constructs a global map of the graph’s connectivity in embedding space (the phenomenon characterized by [3]). Geometric matching is conditional on the *absence of strong interference*.

When interference is weak or absent (uncorrelated words on any graph structure, ordered numbers on circles and grids, days on trees) both modes succeed simultaneously. Loss and energy decrease together, and it is natural to conclude, as [3] did, that geometric reorganization underlies task performance. Our contribution is to show that this coupling is possible, but not necessary. Under strong interference, the two modes do not sync: pattern matching proceeds while geometric matching is blocked. The model reaches perfect accuracy through local statistical learning alone, and the Dirichlet energy remains high.

As a final remark to this decoupling of ICL capabilities, we want to focus on the shuffled numbers on a circle condition. In this setting, the model passes through three phases: at first no circular structure is visible, then, for a brief window around context length = 1480 a clear circle appears in PC1-PC2 and the Dirichlet energy reaches a minimum, finally the energy rises again and the circular structure is lost, even as the loss continues to decrease. We argue that this dynamics reinforces the two-mode picture: pattern matching improves monotonically with context, as one would expect from accumulating

co-occurrence statistics. Geometric matching, by contrast, is fragile: it can appear, stabilize and then collapse as information is redistributed in different direction of variance. Hence the two modes not only can decouple, they can decouple mid-trajectory.

References

- [1] Joshua Engels et al. “Not all language model features are one-dimensionally linear”. In: *arXiv preprint arXiv:2405.14860* (2024).
- [2] Catherine Olsson et al. *In-context Learning and Induction Heads*. 2022. arXiv: [2209.11895](https://arxiv.org/abs/2209.11895) [cs.LG]. URL: <https://arxiv.org/abs/2209.11895>.
- [3] Core Francisco Park et al. *ICLR: In-Context Learning of Representations*. 2025. arXiv: [2501.00070](https://arxiv.org/abs/2501.00070) [cs.CL]. URL: <https://arxiv.org/abs/2501.00070>.